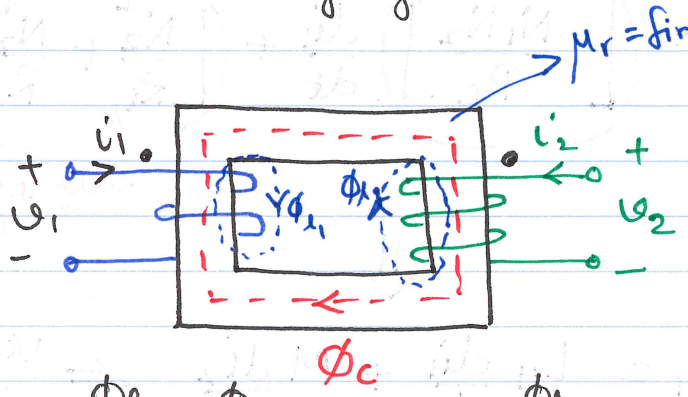
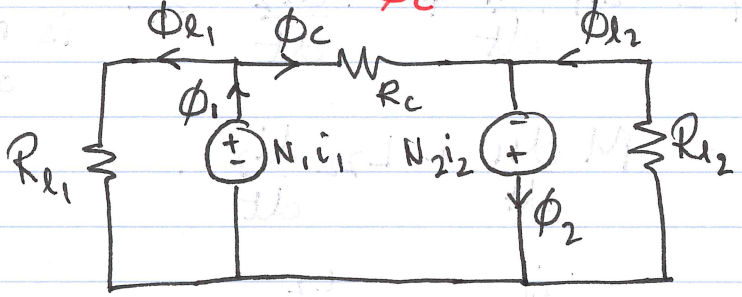
Lecture 12.

Last time: (1) Dot convention
(2) Consideration for $\mu_r \neq \infty$
leakage flux.



Physical model.



Magnetic Equivalent Circuit Model.

$$v_1 = \frac{d\lambda_1}{dt} = N_1 \frac{d}{dt}\phi_1 = N_1 \frac{d}{dt}(\phi_{l_1} + \phi_c)$$

$$= N_1 \frac{d}{dt}\left[\frac{N_1 i_1}{R_{l_1}} + \frac{N_1 i_1 + N_2 i_2}{R_c}\right]$$

$$= \left[\frac{N_1^2}{R_{l_1}} + \frac{N_1^2}{R_c}\right]\frac{d}{dt} i_1 + \frac{N_1 N_2}{R_c}\frac{di_2}{dt}$$

Similarly, $v_2 = \frac{N_1 N_2}{R_c}\frac{di_1}{dt} + \left[\frac{N_2^2}{R_{l_2}} + \frac{N_2^2}{R_c}\right]\frac{di_2}{dt}$.

Self inductance of coil 1 (L_{11})

Mutual inductance (M)

$$v_1 = \left[\frac{N_1^2}{R_{l_1}} + \frac{N_1^2}{R_c}\right]\frac{di_1}{dt} + \frac{N_1 N_2}{R_c}\frac{di_2}{dt}$$

$$v_2 = \left[\frac{N_1 N_2}{R_c}\right]\frac{di_1}{dt} + \left[\frac{N_2^2}{R_{l_2}} + \frac{N_2^2}{R_c}\right]\frac{di_2}{dt}$$

Self inductance of coil 2 (L_{22})

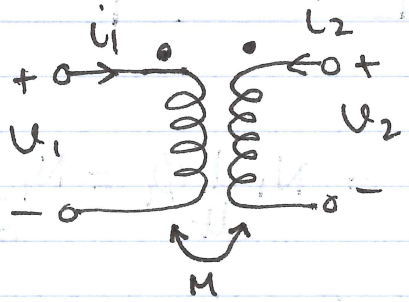
$\Downarrow$

Mathematical model

$$v_1 = L_{11}\frac{di_1}{dt} + M\frac{di_2}{dt}$$

$\Leftarrow$ Voltage in coil 1 is now a function of both coil 1 and 2 current

$$v_2 = M\frac{di_1}{dt} + L_{22}\frac{di_2}{dt}$$



If we have "loosely" coupled coils, M is low.

If we have strongly coupled coils, M is high.

Coefficient of coupling, $K = \dfrac{M}{\sqrt{L_{11}L_{22}}}$

If we assume, no leakage, $R_{\ell_1} = R_{\ell_2} \to \infty$

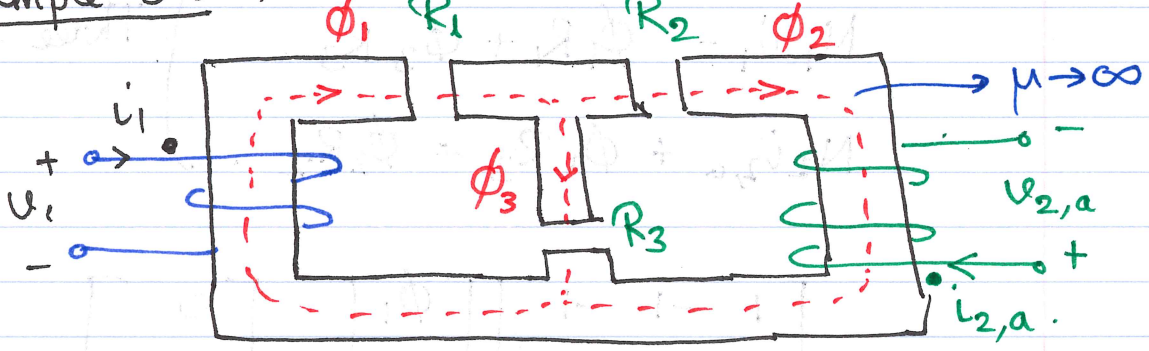
$$\therefore \quad L_{11} = \frac{N_1^2}{R_c} \; ; \; M = \frac{N_1 N_2}{R_c} \; ; \; L_{22} = \frac{N_2^2}{R_c}$$

$$\therefore \quad K = \frac{N_1 N_2 / R_c}{\sqrt{N_1^2 N_2^2 / R_c^2}} = \underline{1} .$$

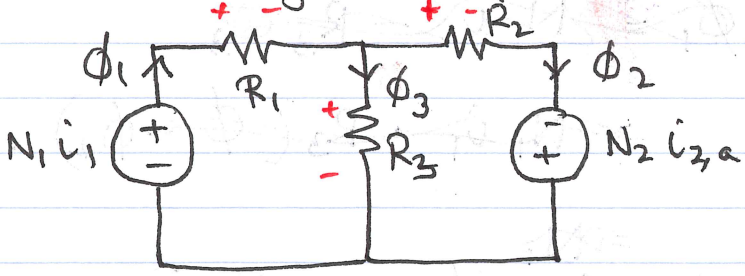
$$\boxed{0 \le K \le 1} \quad \text{Always.}$$

$$\therefore \quad 0 \le M \le \sqrt{L_{11} L_{22}} .$$

Example 3.2 .



Step 1: Put the dot.
Step 2: Draw flux paths.
Step 3: Draw magnetic equivalent circuit.



Flux linkage of wdg #1

$\lambda_1 = N_1 \phi_1$ = must be expressed in terms of i_1 and i_2.

~~$N_1 i_1 + N_2 i_{2,a} = \phi_1 R_1 + \phi_2 R_2$~~.

need to express in terms of i_1 and $i_{2,a}$

~~$N_2 i_{2,a} + \phi_3 R_3 = \phi_2 R_2$~~.

$\phi_1 = \phi_2 + \phi_3$. Three unknowns.

$N_1 i_1 = \phi_1 R_1 + \phi_3 R_3$.

$N_2 i_{2,a} + \phi_3 R_3 = \phi_2 R_2$.

} Three equations

$$\begin{bmatrix} 1 & -1 & -1 \\ R_1 & 0 & R_3 \\ 0 & R_2 & -R_3 \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \end{bmatrix} = \begin{bmatrix} 0 \\ N_1 i_1 \\ N_2 i_{2,a} \end{bmatrix}$$

~~or~~ ~~$R_1 \phi_1 = N_1 i_1 - R_3 \phi_3$~~

~~$N_1 i_1 = R_3 (\phi_1 - \phi_2)$~~.

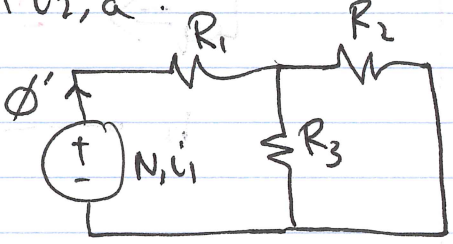
~~=DONE~~

Solve for ϕ_1, that's it.

Is there a better way to solve this? using physical intuition.

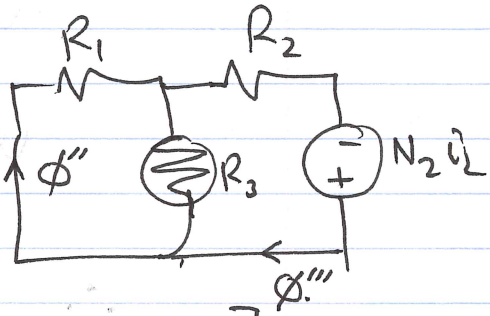
$$\lambda_1 = L_{11} i_1 + M i_{2,a}$$

Ignore wdg #2 current ($i_{2,a} = 0$)



$$L_{11} = \frac{N_1^2}{R_1 + (R_2 \| R_3)}$$

Ignore wdg #1 current, $i_1 = 0$



$$\phi'' = \frac{R_3}{R_1 + R_3} \cdot \frac{N_2 i_{2,a}}{R_2 + (R_1 \| R_3)}$$

$$\therefore \quad M = N_1 N_2 \left[\frac{R_3}{(R_1 + R_3)} \cdot \frac{(R_2 R_1 + R_2 R_3 + R_1 R_3)}{R_1 + R_3} \right]$$

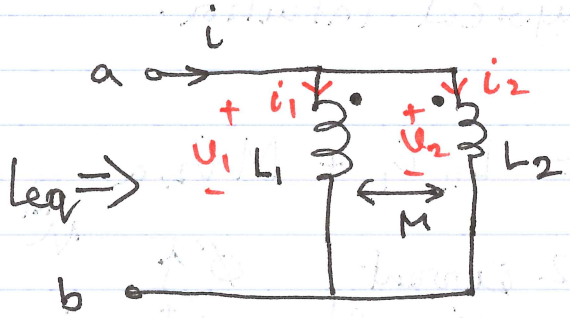
$$= \frac{N_1 N_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}$$

$$\therefore \quad \lambda_1 = \frac{N_1^2 (R_2 + R_3)}{R_1 R_2 + R_2 R_3 + R_3 R_1} i_1 + \frac{N_1 N_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1} i_{2,a}$$

Our, $i_{2a} = -i_{2b}$ ← given in problem.

$$= 3.125 \times 10^{-3} i_1 - 0.9375 \times 10^{-3} i_2$$

Example 3.9:



if $M=0$

$$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}.$$

(parallel inductance).

Coupling makes it interesting.

$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

$\rightarrow v_1 = v_2$

$$\therefore \quad (L_1 - M)\frac{di_1}{dt} = (L_2 - M)\frac{di_2}{dt}.$$

Also, $\quad i = i_1 + i_2$

$$\therefore \quad v_1 = L_{eq} \cdot \frac{di}{dt} = L_{eq} \cdot \frac{d}{dt}(i_1 + i_2).$$

$$L_1 \frac{di_1}{dt} + M \frac{di_2}{dt} = L_{eq}\frac{di_1}{dt} + L_{eq}\frac{di_2}{dt}.$$

$$(L_1 - L_{eq})\frac{di_1}{dt} = (L_{eq} - M)\frac{di_2}{dt}.$$

$$\therefore \quad \frac{L_1 - M}{L_1 - L_{eq}} = \frac{L_2 - M}{L_{eq} - M}.$$

$$(L_1 - M)L_{eq} - \cancel{ML_1} + M^2 = L_1L_2 - \cancel{L_1M} + (L_2 - M)L_{eq}$$

$$\text{or, } \{(L_1 - M) + (L_2 - M)\}L_{eq} = L_1L_2 - M^2$$

$$\therefore \quad L_{eq} = \frac{L_1L_2 - M^2}{L_1 + L_2 - 2M}.$$